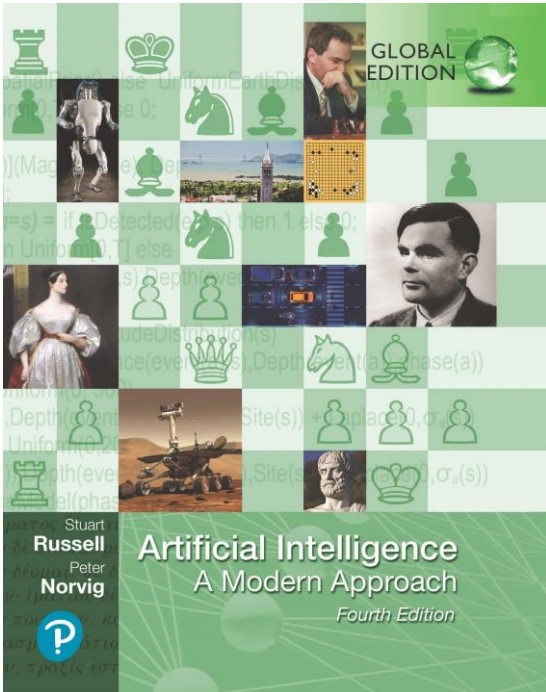


Chapter 3

Solving Problems by Searching

Homework # 3

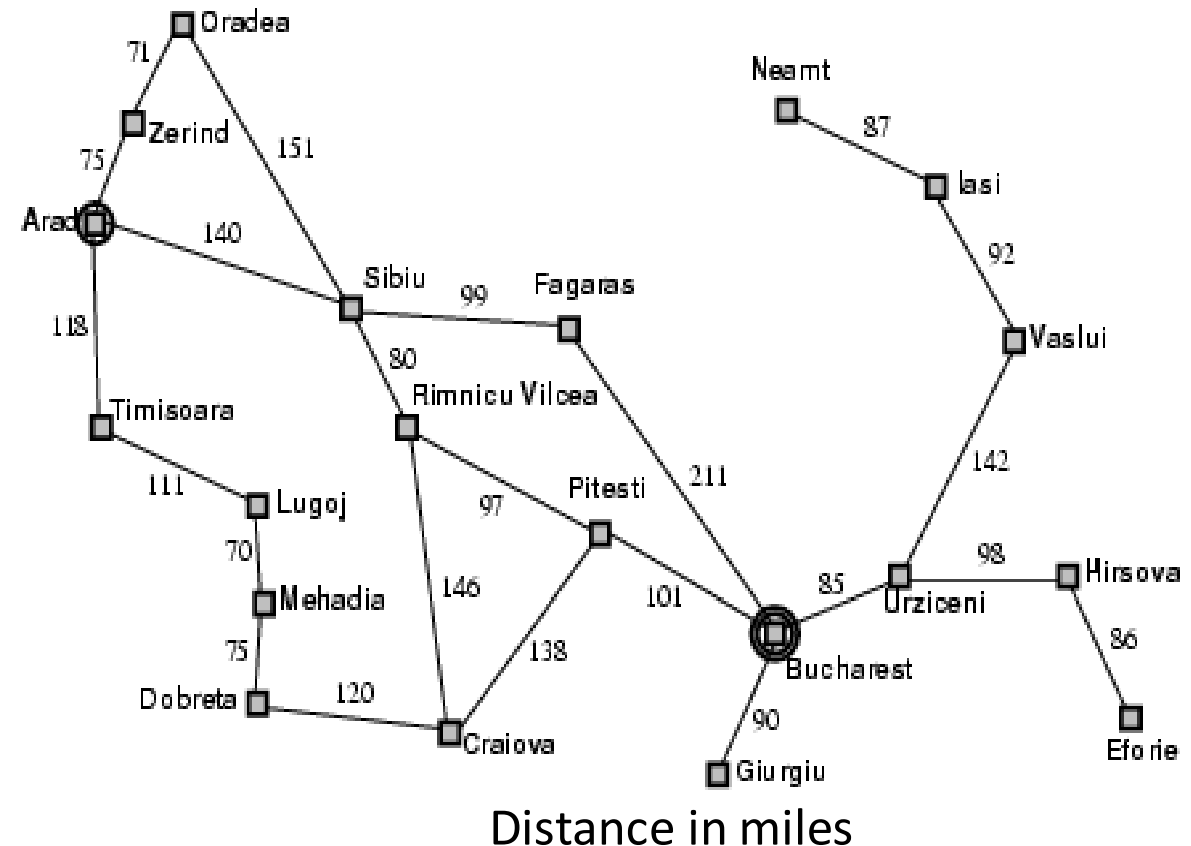


Implementation of the trip in Romania (ARAD-Bucharest)

- Write a Python program to implement the Romania's trip Intelligent agent where **A* search** is defined using Python.
 - Program should be flexible to select any state as initial state, find the lowest cost path to reach the final state.
 - Use the definitions on the next page.

- Important definitions

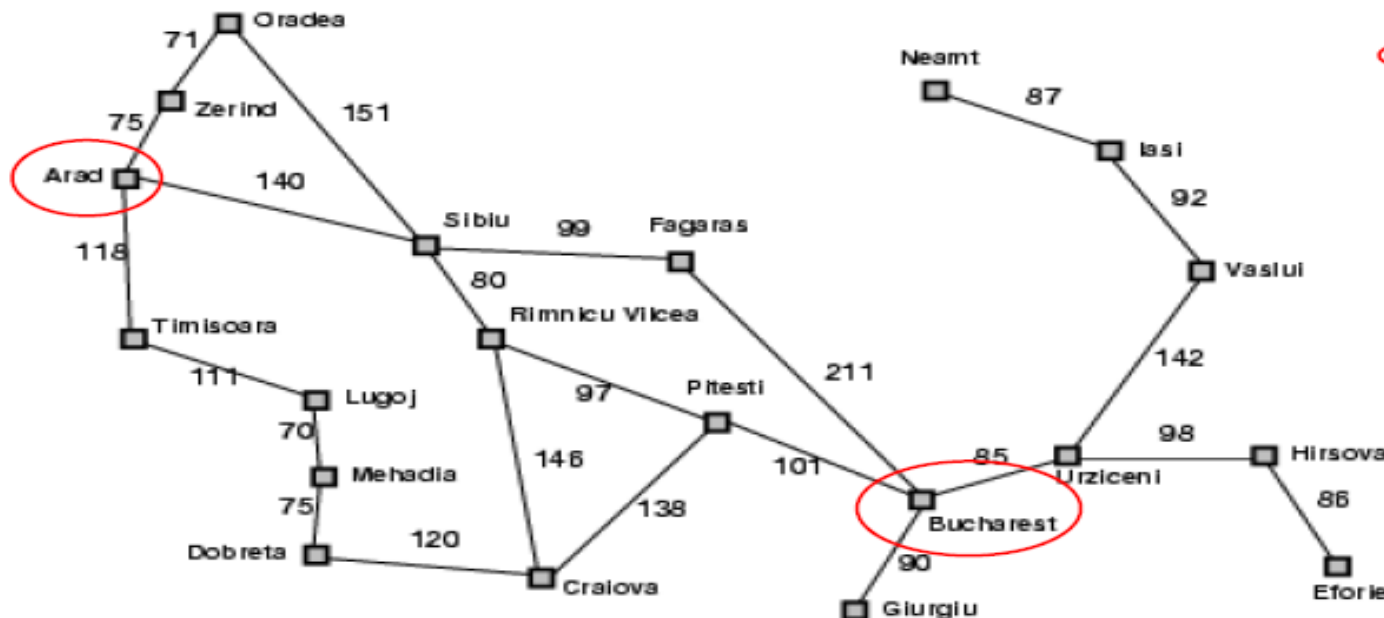
- **Initial state:** Arad
- **Actions:** Drive from one city to another
- **Transition model and states:** If you go from city A to city B, you end up in city B
- **Goal state:** Bucharest
- **Path cost:** Sum of edge costs



Implementation of the trip in Romania (ARAD-Bucharest)

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 - Program should be flexible to select any state as initial state, find the lowest cost path to reach the final state.
 - Use the definitions on the next page.
 - $g(n)$ is distance from origin (Arad) (see lecture)

$$f(n) = g(n) + h(n) = \text{Arad} \rightarrow 366 = 0 + 366$$



$h(n)$

Straight-line distance
to Bucharest

Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	176
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	101
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374